01 / PROBLEM AND CONTRIBUTION

Audio encryption/decryption study

Encryption / decryption case study | Undergraduate | Three-person study

RESULT AT A GLANCE

Presented original, encrypted, and recovered signal views in an educational MATLAB study, with the scope limited to the documented workflow rather than a security benchmark.

The engineering question

How can an audio signal illustrate the distinction between encryption and decryption?

Situation

An audio signal provides a visual example of an encryption/decryption sequence. This team study discusses symmetric and asymmetric concepts with RSA as its reference algorithm.

Task

Contribute as a team member to the conceptual study and documented MATLAB signal workflow.

My contribution

Listed team member in the conceptual study and MATLAB demonstration.

Tools and concepts

MATLAB / Audio processing / RSA concepts / Algorithms / Technical documentation

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

Explored symmetric and asymmetric key concepts and used RSA as the report's reference algorithm.

02 / Implementation step

Documented a MATLAB workflow that starts with an audio file, illustrates an encrypted signal, and shows a recovered signal at the receiving end.

Educational signal workflow

The report uses original, encrypted, and recovered signal views to explain the processing sequence. Symmetric/asymmetric concepts and RSA are part of its discussion; those descriptions do not independently verify implementation security.

Architecture / workflow

Conceptual summary of the documented workflow; not a simulation result.

01	Audio input
02	Encryption step
03	Encoded audio
04	Decryption step

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

Reported result

The document contains example signal views and a conceptual discussion of the encryption/decryption sequence.

Recovery and security are separate

A recognizable recovered signal is an illustration of the workflow. Quantitative fidelity, key handling, and cryptographic correctness would need separate evidence before describing the example as a secure audio system.

Scope and limitations

This is an educational algorithm study. The signal illustrations explain the workflow; cryptographic validation and quantitative recovery analysis sit beyond its demonstrated scope.

Possible next steps - proposed, not completed results

- Document the exact algorithm, key handling, and reproducible input/output procedure.
- Measure recovery error and seek an appropriate cryptographic review before making security claims.

Source material: REAL TIME APPLICATION.docx in gptz.zip, case-study and team sections.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.